

# Skills, Inequality, and Referrals\*

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December 13, 2024

## 1 Background and Setting

*Inequality and Social Class.* Colombia has consistently ranked as one of the most unequal countries in Latin America (World Bank, 2024), with the richest decile earning 50 times more than the poorest decile (ECLAC, 2023). This economic disparity is reflected by a highly stratified society, where social classes remain separated by education, family background, lifestyle, occupation, power, and geographic residence.

*Stratum system.* In 1994, Colombia introduced a nationwide classification system dividing the population into 6 strata based on housing characteristics and neighborhood amenities. This system, initially designed for utility subsidies from higher strata (5 and 6) to support lower strata (1 to 3), now extends to university fees and social program eligibility. While imperfect -because it disregards the income of residents, this system largely aligns with and potentially reinforces existing social class divisions (Guevara S & Shields, 2019; Uribe-Mallarino, 2008). We use this exogenous cutoff as the measure of social class in our experiment: Students in strata 1 to 3 are categorized as low-SES,

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\*We obtained Institutional Review Board approvals from NYU Abu Dhabi (HRPP 2024-50) and the University of Luxembourg (ERP 24-028). The study design was preregistered in the OSF Registries prior to data collection (see <https://doi.org/10.17605/OSF.IO/V9T3W>).

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and those in strata 4 to 6 as high-SES.

*Universidad Autónoma de Bucaramanga.* Our study takes place at UNAB, a medium-sized private university in Bucaramanga, Colombia, with approximately 10,000 enrolled students. UNAB’s student body is notably diverse, with slightly more than half of the students classified as low-SES (see Appendix Figure A.1 for a detailed stratum distribution). This diversity reflects the broader trend in Colombian society, where even lower middle-class families (stratum 2 and 3) prioritize university education for their children despite the financial burden (Hudson & Library of Congress, 2010, p. 103). The university environment provides a unique opportunity for collaborative inter-class contact on equal status, whose positive effects on reducing discrimination are casually documented (Lowe, 2021; Mousa, 2020; Rao, 2019). We draw our sample from a cohort of students enrolled in two semester-long courses. Each course meets weekly for three-hour sessions. Attendance is mandatory, but enforcement varies across instructors. Both courses are university-wide graduation requirements which result in large variations in academic programs and socioeconomic backgrounds across the classrooms.

## 2 Design

We designed a referral experiment between students from different social classes, and experimentally varied the incentives to refer a low-SES peer. The study design consists of a single experiment with three sequential parts with sessions organized at the classroom level (see Figure 1). The instructions are provided in Appendix B.

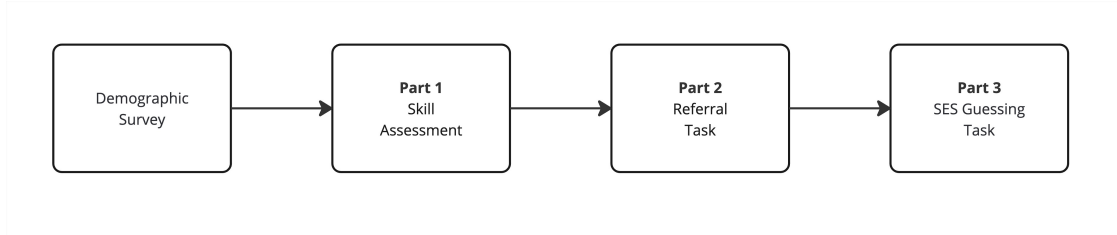
### 2.1 Skill Assessment

To understand the basis for referral decisions, we collect objective measures of cognitive and social skills. These two distinct skills are crucial for the labor market and suitable in a context with university students.<sup>1</sup> This approach ensures that we observe the distri-

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<sup>1</sup>For the relationship between wages and social skills conditional on cognitive skills, see Deming (2017); Edin, Fredriksson, Nybom, and Öckert (2022); Kuhn and Weinberger (2005); Weinberger (2014).

Figure 1: Experiment Timeline



*Note:* Participants first complete incentivized skill tests, then refer classmates for skills. In the final part, they guess the social class of their peers. This order is implemented in all sessions.

bution of skills in our sample, ask for referrals on these skills, and compare how well the same individuals refer peers for a specific skill. Participants perform two incentivized skill tests. They have 5 minutes to complete each test. We provide test-specific instructions and an example item before participants begin. Correctly solved items increase chances to earn a fixed bonus.<sup>2</sup>

*Cognitive Skills.* We use Raven’s Progressive Matrices to measure cognitive skills (Raven, 1936; Raven, Raven, & Court, 1976). Raven’s test is a well-established measure of fluid intelligence, i.e., an individual’s capacity to reason and solve problems in novel situations independent of past knowledge (Schilbach, Schofield, & Mullainathan, 2016). In this test, participants see series of images where there is a pattern with a piece that has been intentionally removed. They are tasked with choosing the piece that completes the pattern among available options. For each image, there is only one correct answer. We implement an 18-item version featuring increasingly difficult questions, with 6 response options for the first 9 items and 8 thereafter.

*Social Skills.* We measure social skills with the Multiracial Reading the Mind in the

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<sup>2</sup>The tests are presented in a randomized order. No performance feedback is provided. Participants see one item at a time and cannot return to previous screens once they start a test. They are not required to answer items and can skip them if they choose to do so. We elicit beliefs about performance after each test.

Eyes Test (MRMET) from Kim et al. (2022).<sup>3</sup> The test is an established measure for the ability to recognize emotions in others, and it has been previously used in economic experiments (van Leeuwen et al., 2018; Weidmann & Deming, 2021). MRMET consists of photos of human faces portraying different emotions, cropped so that only the eye region is visible. Participants must choose the emotion that best describes the photo from the available answers. For each photo, there is only one correct answer and 4 response options. We administer the first 36 items in MRMET.

## 2.2 Referral Task

We create a referral task to measure bias against low-SES peers. For each skill, participants make incentivized referrals by nominating classmates. We first explain the measured skill accompanied by an example test item. On the following page, we provide an alphabetically ordered list of all classmates. Participants make three referral choices per skill. They are instructed to exclude themselves from referrals. A classmate may be nominated once per triad. The order in which participants refer for a skill test is randomized. We incentivize referrals with classroom-level performance rankings. The three highest-scoring classmates are designated as the top 3 for a skill. Referrers are eligible for a fixed bonus for referrals among the top 3.<sup>4</sup>

*Treatment.* We have one between subject treatment, and it varies the top 3 selection. In the **Baseline** treatment, the top 3 selection is based solely on performance ranking, regardless of other participant characteristics. The **Quota** treatment modifies the top 3 selection to prioritize low-SES individuals. We reserve the first spot in the top 3 for the highest-scoring low-SES classmate, and assign the remaining two places based on performance. This guarantees at least one low-SES participant in the top 3 per skill. Participants are informed about the top 3 selection mechanism before making referral

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<sup>3</sup>We choose MRMET because it is a race- and gender-inclusive test suitable for application in non-WEIRD (Western, Educated, Industrial, Rich, Democratic) populations like the one we sample from. The test is based on the original RMET (Baron-Cohen, Wheelwright, Hill, Raste, & Plumb, 2001).

<sup>4</sup>We solve ties among the top 3 randomly. We describe only the top 3 selection mechanism and provide no feedback about the top 3 composition to participants.

choices (Appendix Figure A.7 provides illustrations explaining the treatments). Assignment to the treatment is at the individual level within each classroom. This allows comparing the effect of the treatment while keeping the referral choice set constant.

### 2.3 SES Guessing Task

Participants make guesses about the anticipated SES of their classmates. We inform participants that a computer algorithm randomly selects three students belonging to strata 1, 2, or 3. They are tasked with nominating the people they believe the computer could choose at random (Appendix Figure A.8 provides the illustration explaining the task). Participants select three classmates from an alphabetically ordered list containing all their classmates. This task measures the ability to distinguish-SES independent of test performance, as SES identification is relevant to our study.

## 3 Sample, Incentives, and Procedure

*Sample.* We invited 849 UNAB undergraduate students to participate in the experiment. Our final sample consists of 713 individuals who completed the study protocol, resulting in an 83% participation rate. Table 1 presents key demographic characteristics and academic performance indicators across treatments (Appendix Table A.1 compares the sample and missing students). The sample is well-balanced between the Baseline and Quota conditions. We observe no statistically significant differences in any of the reported variables (all  $p$  values  $> 0.1$ ), indicating balanced randomization. The sample is characterized by a majority of low-SES students. About one-third of the sample are first-generation college students. The gender distribution is balanced. The mean GPA of 3.95 is consistent across both treatments.

*Incentives.* Participants could earn bonuses worth 100,000 Pesos (about 26 US Dollars) in each part of the experiment. In the first part, we incentivized performance in the skill tests. 20% of participants were eligible for the bonus. We randomly picked one skill test for each eligible participant and drew a number between 1 and 100. The participant

received the bonus if the percentage of correct answers in the selected test exceeded the drawn number. Chances of earning the bonus increased with each correctly solved question by 5.5% ( $=1/18$ ) for the Cognitive Skill test and by 2.78% ( $=1/36$ ) for the Social Skill test. In the second part, we incentivized referrals among the top 3 performers. 40% of participants were eligible for the bonus. We randomly selected one skill test and one referral for each eligible participant. The participant received the bonus if their referral was among the top 3. In the third part, we incentivized the correct identification of low-SES classmates. 20% of participants in each classroom were eligible for the bonus. We randomly selected one guess for each eligible participant. The participant received the bonus if their guess correctly identified a low-SES classmate.

*Procedure.* Data collection occurred during the last weeks of April 2024. Our local partner, the Social Bee Lab, coordinated scheduled classroom visits and recruited research assistants to administer the experiment. Students present in class on the scheduled visit dates participated. Each classroom visit constituted a separate session. There were in total 35 sessions. Participants accessed the Qualtrics-based experiment using their smartphones during these visits. The median time to complete the survey was 20 minutes, with a compensation of \$26 for 117 lottery winners.

Table 1: Comparison of demographic characteristics and academic outcomes across treatment conditions

|                             | <b>Baseline</b> | <b>Quota</b> | <b><i>p</i></b> |
|-----------------------------|-----------------|--------------|-----------------|
| Low-SES                     | 59%             | 55%          | 0.297           |
| Cognitive skill (out of 18) | 10.04           | 10.27        | 0.322           |
| Social skill (out of 36)    | 18.45           | 18.50        | 0.886           |
| GPA                         | 3.95            | 3.95         | 0.828           |
| Entry Exam                  | 61.85           | 62.17        | 0.638           |
| Age                         | 19.33           | 19.02        | 0.228           |
| Female                      | 52%             | 47%          | 0.195           |
| First Generation            | 34%             | 37%          | 0.386           |
| Ethnic Minority             | 1%              | 3%           | 0.133           |
| Rural Community             | 30%             | 27%          | 0.308           |
| Has Scholarship             | 1%              | 1%           | 0.916           |
| # Semesters at UNAB         | 3.18            | 3.17         | 0.916           |

*Note:* Low-SES, female, first generation, ethnic minority, rural, and has scholarship represent binary outcomes and *p*-values for these variables are from two-sample tests of proportions. For continuous variables (age, GPA, cognitive skill, social skill, entry exam, and semester), *p*-values are from two-sample t-tests with equal variances. All reported *p*-values are two-tailed. Sample sizes vary slightly across measures due to missing data.

## 4 Empirical Analysis

### Can peers identify skills?

Our first goal is understanding whether higher skilled individuals get more referrals. Because every referrer nominates 3 classmates per skill, analyzing only the extensive margin, i.e., whether an individual gets a referral, is not very informative.<sup>5</sup> We consider the percentage share of referrals from individuals in Baseline condition as our dependent variable. This approach combines the intensive and extensive margins and also makes comparisons across classrooms with different sizes easier.<sup>6</sup> Formally, we define the percentage share of referrals received by individual  $i$  from participants in classroom  $c$  and in Baseline condition ( $\forall j \in B_c$ ) for skill  $s \in \{Cognitive, Social\}$  as:

$$y_{ic}^s = \frac{\sum_{j \neq i} r_{ijc}^s}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (1)$$

where  $n_c$  represents the number of participants in the Baseline condition in classroom  $c$ . The indicator  $r_{ijc}^s$  takes value 1 if participant  $j$  in the Baseline condition refers individual  $i$  for skill  $s$ , and 0 otherwise, and require both  $i$  and  $j$  to be in the same classroom  $c$ . The denominator  $n_c - \mathbb{1}(i \in B_c)$  accounts for the maximum possible referrals that individual  $i$  could receive. If  $i$  is in the Baseline condition ( $\mathbb{1}(i \in B_c) = 1$ ), we subtract one from  $n_c$  to account for the self-referral restriction.<sup>7</sup> This normalized measure represents the percentage of potential referrals actually received by each individual, adjusting for classroom size and treatment status. By construction,  $y_i^s \in [0, 100]$  for all  $c$ , and we can compare referrals across classrooms of different sizes. Appendix Figure

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<sup>5</sup>Only 86 of the 849 students (10%) never get a referral for either skill.

<sup>6</sup>The number of participants in a classroom mechanically drives the number of total referrals that could be received by an individual. By normalizing referrals we focus on differences within classrooms.

<sup>7</sup>We find that 33.8 percent of participants in the sample for cognitive and social skills self-referred, while explicitly instructed not to do so. In Appendix Table A.4 we compare the outcomes of the subsample who self-refer. Those who self-refer are 10 percent more likely to be low-SES, and have systematically lower cognitive skills (0.2 SD) and GPA (0.25 SD). As self-referrers are not informative and add noise to our estimates, we drop these instances from our paired referral-referrer sample in subsequent analyses. Self-referring participants' remaining referral choices are kept in the dataset.

A.2 presents the distribution of our dependent variables and illustrates with an example. Under performance pay, higher performing classmates in skill tests should collect more referrals if classmates can successfully identify skills. Our only independent variables are standardized skill test scores. We estimate referral percentage shares  $y_i^s$ :

$$y_i^s = \beta_0^s + \beta_1^s Skill_i^s + \epsilon_i^s \quad (2)$$

Table 2 illustrates our first findings. The comparison of interest is the point estimates for different skills. In column (1), a one standard deviation increase in cognitive skill causes a 1.2 percentage point increase in the share of referrals received. On a base rate of 13%, this is a modest increase of 9 percent. In column (2), 95% confidence intervals rule out that a one standard deviation increase in the social skill result in more than a 1 percentage point difference in the share of referrals received. Participants have difficulties identifying skills in the Baseline condition, with modest recognition of cognitive and no recognition at all of social skills.

Table 2: Share of referrals conditional on skill

|              | (1)                  | (2)                  |
|--------------|----------------------|----------------------|
|              | Cognitive            | Social               |
| Skill        | 1.197**<br>(0.479)   | 0.037<br>(0.510)     |
| Constant     | 12.986***<br>(0.653) | 13.049***<br>(0.521) |
| Observations | 665                  | 665                  |

*Note:* Classroom-level clustered standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Dependent variables are the percentage of referrals received relative to all referrals. “Skill” refers to standardized test scores for cognitive and social skills. Sample restricted 665 individuals for whom we have complete administrative and experimental data.

In the absence of a clean skill-signal from their peers, participants may have referred

classmates using proxies of skills. We define proxies as peer beliefs about strong correlates for skills. A straightforward proxy for cognitive skills would be the “good students” in the classroom, as measured by GPA. However, cognitive skills and grades are far from perfectly correlated (Heckman & Kautz, 2012), and such beliefs may not be accurate. Indeed, in our sample, GPA correlates very weakly with skills (see Appendix Table A.2). We attempt to capture these beliefs by including standardized GPA scores as an independent variable in Equation (2) and re-estimate referral percentage shares  $y_i^s$ .

Table 3 illustrates our findings. The comparison of interest is the difference between point estimates for skills and GPA. In column (1), a one standard deviation increase in cognitive skill causes a 0.9 percentage point increase in the share of referrals received when controlling for GPA. On a base rate of 12.8%, this is a comparable increase in magnitude of about 7 percent to our previous estimate in Table 2, and suggests cognitive skills have an independent effect on referrals. However, a one standard deviation increase in GPA causes a substantial 3.9 percentage point increase in the share of referrals received when controlling for cognitive skill. This is an increase of more than four times in terms of magnitude (30 percent) when compared to cognitive skill, and suggestive of the extent to which academic performance is easier to observe among peers. In column (2), 95% confidence intervals rule out that a one standard deviation increase in the social skill result in more than a 1 percentage point difference in the share of referrals received. This finding is consistent with our previous estimate where participants cannot identify social skills. On the other hand, a one standard deviation increase in GPA causes a substantial 3.4 percentage point increase in the share of referrals received when controlling for social skill. This is a similar albeit slightly smaller increase of 26 percent in the share of referrals for social skill. For both skills, we find strong evidence that grades act as a proxy for referral decisions. In the subsequent section we expand more on how referral decisions are made.

Table 3: Share of referrals conditional on skill and academic performance

|              | (1)       | (2)       |
|--------------|-----------|-----------|
|              | Cognitive | Social    |
| Skill        | 0.873*    | -0.278    |
|              | (0.467)   | (0.460)   |
| GPA          | 3.949***  | 3.429***  |
|              | (0.664)   | (0.581)   |
| Constant     | 12.806*** | 12.891*** |
|              | (0.662)   | (0.636)   |
| Observations | 665       | 665       |

*Note:* Classroom-level clustered standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Dependent variables are the percentage of referrals received relative to all referrals. “Skill” refers to standardized test scores for cognitive and social skills. GPA is standardized to mean zero and unit variance.

### Overlapping referrals

Diversity in referral choices vary substantially between participants as they nominate 3 individuals per skill and can make overlapping referrals across skills. In Appendix Figures A.3a and A.3b, we show that only 20.5 percent of participants pick all unique classmates for their 6 choices. In fact, depending on our measures, between 50 to 70 percent of participants make at least 2 overlapping referrals and the diversity in referral choices is much lower than anticipated.<sup>8</sup>

With such a large share overlapping referral choices, it is possible that participants believed classmates with a higher score in one skill would also have a higher score in the other. Would such beliefs be accurate? There is only modest ( $\rho = 0.267$ ) correlation between the two skills (see Appendix Table A.2). To understand whether making over-

<sup>8</sup>In Appendix Table A.5 we compare the characteristics of referrers make unique referrals to the remainder of the sample. The results suggest minimal differences in GPA and skills.

lapping referrals is strategic, we turn to the incentives. Participants were incentivized to pick the top 3 performers in their classroom for each skill to earn a fixed bonus. Looking at the characteristics of top skilled participants in Appendix Table A.6, we find that conditional on being among the top 3 for one of the skills, only 1 in 3 participants were in the top 3 for both skills. This suggests making more than 1 overlapping referral choice would decrease the chances to win the bonus.

A competing explanation for the amount of overlapping referral choices coupled with the notable difficulties in identifying skills would be that individuals who refer classmates twice for both skills are worse at observing peer skills. This implies the underlying heterogeneity in skill identification results in differential referral strategies where participants with a good signal for a skill choose to refer classmates only once for that skill, and those without a good signal use the grade proxy and refer classmates for both skills. We can test both hypotheses in our data: If “twice” referrers -defined as those who refer an individual for both skills- are better at identifying at least one of the skills, this gives more credence to beliefs about correlated skills. On the other hand, if “twice” referrers are worse in skill identification compared to single-referrers and use GPA proxy for referrals, they have no information about skills.

We compare the outcomes of participants who receive referrals twice from their classmates to those who receive referrals only once per skill. Formally, let indicator  $r_{ijc}^{twice}$  take value 1 if individual  $j$  referred individual  $i$  for both skills. The percentage share of referrals received by individual  $i$  from participants in classroom  $c$  and in Baseline condition ( $\forall j \in B_c$ ) is:

$$y_{ic}^{twice} = \frac{\sum_{j \neq i} r_{ijc}^{twice}}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (3)$$

where  $n_c$  represents the number of participants in the Baseline condition in classroom  $c$ . The indicator  $r_{ijc}^{twice}$  takes value 0 if participant  $j$  in the Baseline condition does not refer individual  $i$  for both skills. The denominator  $n_c - \mathbb{1}(i \in B_c)$  accounts for the maximum possible “twice” referrals that individual  $i$  could receive as before. Similarly, let  $r_{ijc}^{s, single}$  take value 1 if individual  $j$  referred individual  $i$  only for skill  $s$ . The percentage

share of “single” referrals received by individual  $i$  from participants in classroom  $c$  and in Baseline condition ( $\forall j \in B_c$ ) for skill  $s$  is:

$$y_{ic}^{s, \text{single}} = \frac{\sum_{j \neq i} r_{ijc}^{s, \text{single}}}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (4)$$

and it follows that for any  $s$ , percentage share of “single” and “twice” referrals received by individual  $i$  from participants in classroom  $c$  and in Baseline condition ( $\forall j \in B_c$ ) must add up to the total share of referrals received:

$$y_{ic}^s = y_{ic}^{s, \text{single}} + y_{ic}^{\text{twice}} \quad (5)$$

we regress Equation 2 supplanted with GPA for our three new dependent variables and report our findings in Table 4. The comparison of interest is the skill and GPA estimates across columns. In column (1), we find that a one standard deviation increase in GPA causes a 3.2 percentage point increase in the share of “twice” referrals received when controlling for skills. This is a staggering 43 percent increase on a base rate of 7.4%. 95% confidence intervals rule out that a one standard deviation increase in either skill result in more than a 1 percentage point difference in the share of “twice” referrals received. Participants who nominate the same individuals for both skills make referrals based on academic performance and cannot observe classmates with higher cognitive or social skills. For participants who receive only cognitive skill referrals, in column (2), we find that a one standard deviation increase in GPA causes a 0.8 percentage point increase in the share of referrals when controlling for cognitive skill. A one standard deviation increase in cognitive skill causes a comparable 1 percentage point increase in referrals when controlling for GPA. These are respectively 15 and 19 percent increases in the share of referrals received, and suggest participants are able to identify higher skilled peers when referring only for cognitive skill. The lower base rate of 5.4% compared to 7.4% in column (1) suggest less than half of referrals came from “single” referrals. GPA estimate is three times smaller in magnitude compared to column (1), and is supportive of the smaller weight put in the grades proxy. Nevertheless, the similar magnitudes of GPA and cognitive skill point estimates still suggest participants refer peers with higher

grades much more often than the correlation between the two supported by the data ( $\rho = 0.085$ ). There is heterogeneity in skill identification even for those who refer a peer once for cognitive skill. For participants who receive social skill referrals only, in column (3), 95% confidence intervals rule out that a one standard deviation increase in social skill or GPA result in more than a 1 percentage point difference in the share referrals received. These further support our previous finding that peers cannot identify social skills in our sample, and attempts to identify social skills by ceasing the use of the GPA proxy seems to fail. Taken together, we find that most individuals cannot identify skills in our sample, instead refer using the GPA proxy, and predominantly refer the same individuals for both skills. Those who refer an individual only once for cognitive skill drive the entirety of the results in terms of peer skill recognition. Yet, even these individuals confound the cognitive skill signal with academic performance, and put similar weights on the two characteristic.

Table 4: “Twice” versus “single” referral share conditional on skills and academic performance

|                 | (1)                 | (2)                 | (3)                 |
|-----------------|---------------------|---------------------|---------------------|
|                 | Twice               | Cognitive Only      | Social Only         |
| GPA             | 3.172***<br>(0.464) | 0.801**<br>(0.391)  | 0.260<br>(0.334)    |
| Cognitive Skill | -0.042<br>(0.416)   | 1.006***<br>(0.270) |                     |
| Social Skill    | -0.353<br>(0.304)   |                     | 0.086<br>(0.381)    |
| Constant        | 7.407***<br>(0.524) | 5.400***<br>(0.386) | 5.485***<br>(0.376) |
| Observations    | 665                 | 665                 | 665                 |

*Note:* Classroom-level clustered standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable in column (1) is the percentage share of “twice” referrals received from the same referrer, in column (2) “single” referral share for cognitive skill, and in column (3) for social skill. Independent variables are the respective standardized test scores for skills and GPA.

## Quota Treatment and Low-SES

In the following empirical specification, we document whether there is a social class bias in referrals, and whether the Quota treatment causes referral shares of low-SES participants to change when controlling for skills and academic performance. We hypothesize that the Quota treatment should increase referrals to low-SES classmates because of the additional incentive to refer low-SES. Our dependent variable is the percentage share of referrals received as defined for the Baseline treatment in Equation 2, and extend to the Quota treatment. It is trivial to see  $y_{ic}^s$  can also be calculated for the Quota treatment as participants in every classroom are randomized into either treatment. Now, every partic-

ipant is observed twice in the data, and we add a treatment dummy to indicate whether the referrals came from participants in the Baseline or the Quota treatment. We also add a social class dummy for the participant receiving the referrals to our specification. We estimate:

$$y_i^s = \beta_0^s + \beta_1^s Quota_i + \beta_2^s SES_i + \beta_3^s (Quota_i \times SES_i) + \beta_4^s Skill_i^s + \beta_5^s GPA_i + \epsilon_i^s \quad (6)$$

Table 5 illustrates our findings. Our comparison of interest is the effect of the Quota treatment on low-SES peers. In column (1) for cognitive skill, we find that being low-SES decreases the share of referrals recieved by about 1.2 percentage points when controlling for the skill and academic performance in the Baseline condition. This goes down to about 1.5 percentage points in the Quota condition. Although these differences are not statistically significant, their sign suggests a relatively large bias against low-SES classmates: A one standard deviation increase in cognitive skill has less than half this magnitude. In column (2) for social skill, we find that being low-SES has no statistically significant effect on the share of referrals recieved when controlling for the skill and academic performance. The sign for the point estimate suggest low-SES get slightly more referrals in the Baseline condition, and the Quota reduces this share. The lack of statistical significance in the SES main effect and interaction term suggests considerable noise in both estimates.

Table 5: Effects of Quota on low-SES controlling for skill and academic performance

|                        | (1)                  | (2)                  |
|------------------------|----------------------|----------------------|
|                        | Cognitive            | Social               |
| Quota                  | -0.073<br>(0.755)    | 0.299<br>(0.716)     |
| Low-SES                | -1.230<br>(1.079)    | 0.364<br>(1.282)     |
| Quota $\times$ Low-SES | -0.167<br>(1.117)    | -0.835<br>(1.181)    |
| Skill                  | 0.594<br>(0.448)     | 0.201<br>(0.426)     |
| GPA                    | 3.184***<br>(0.517)  | 2.819***<br>(0.493)  |
| Constant               | 13.551***<br>(0.986) | 12.706***<br>(0.952) |
| Observations           | 1,330                | 1,330                |

*Note:* Classroom-level clustered standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variables are the percentage share of referrals received for cognitive and social skills. Quota is a dummy for the referrals received from classmates in the Quota treatment. Low-SES is a dummy for participant’s socioeconomic status. Remaining independent variables are the respective standardized test scores for skills and GPA.

It is possible that the effects of social class get diluted in the substantial heterogeneity in referrals. A large proportion of participants -“twice” referrers- struggle with skill identification and use the academic performance proxy. In our sample we find no SES differences for GPA (see Appendix Figure A.4). In fact, past evidence suggests low-SES may even outperform high-SES at the university level in some samples [CITE](#)

**MANU WORK.** When referrals are made with academic performance in mind, there is no particular reason to observe a negative bias against low-SES. How about skills? In our sample high-SES have higher skill scores in both measures (see Appendix Figure A.5a and A.5b). If individuals who refer only cognitive skills have a clean signal about the skill, it may be reasonable to observe a bias in terms of undersampling from high-performing low-SES. Only then would the Quota treatment be effective in increasing referrals for low-SES. We turn toward our classification of different referral strategies to test this hypothesis. To get clearer estimates for the social class bias and the effects of the Quota treatment on low-SES referrals, we re-estimate the shares of “twice” and “single” referrals. Following the same logic as before, we observe every participant twice in each specification, and add a treatment dummy to indicate whether the referrals came from referrers in the Baseline or the Quota treatment. We keep the social class dummy and regress Equation 6 for the three dependent variables.

Table 6 illustrates our findings. The comparison of interest is the SES of the participant receiving the referrals and the effect of the Quota treatment across “twice” and “single” referral types. In column (1), for participants who refer the same peers twice using the academic performance proxy, we find no statistically significant effect of participant SES or the Quota on the referrals share when controlling for skills and academic performance. The signs on the point estimates suggest that directionally low-SES get more referrals in the Baseline treatment and that the Quota reduces this share. For participants who make only cognitive skill referrals, in column (2), we find that being low-SES in the Baseline treatment reduces the percentage share of referrals received by 2.1 percentage points when controlling for the skill and academic performance. This is a very large effect size in magnitude, translating to a decrease in referral share by 32 percent on a base rate of 6.6%. In turn, the Quota treatment increases referrals to low-SES by 1.42 percentage points when controlling for the skill and academic performance. This is also a large effect size that results in an increase in low-SES referral share by 22 percent. For participants who make only social skill referrals, in column (2), we find no statistically significant effect of participant SES or the Quota on the referrals share when

controlling for the skill and academic performance. These are in accordance with our previous findings that social skills cannot be identified and provide no reason to warrant social class bias. Taken together, we find support for our hypothesis that there is a bias against low-SES peers for the skill that is well-identified by peers, and in which low-SES underperform. We find no evidence of a bias when referrals are made based on academic performance where both social classes perform equally well. Reassuringly, we find the bias in cognitive skill referrals is at least partially alleviated by the Quota treatment. Because there is remarkable heterogeneity in the ability to detect SES for both social classes (see Appendix Figure [A.6](#)), we find this increase in low-SES referrals satisfying for our sample. In the next section, we explore the possible origins of the social class bias.

Table 6: Effects of Quota on “twice” and “single” referral share controlling for skill and academic performance

|                        | (1)                 | (2)                  | (3)                 |
|------------------------|---------------------|----------------------|---------------------|
|                        | Twice               | Cognitive Only       | Social Only         |
| Quota                  | 0.436<br>(0.817)    | -0.509<br>(0.598)    | -0.136<br>(0.523)   |
| Low SES                | 0.857<br>(0.920)    | -2.074***<br>(0.722) | -0.510<br>(0.613)   |
| Quota $\times$ Low-SES | -1.584<br>(1.159)   | 1.417**<br>(0.656)   | 0.750<br>(0.717)    |
| Cognitive Skill        | -0.079<br>(0.374)   | 0.658***<br>(0.201)  |                     |
| Social Skill           | 0.062<br>(0.283)    |                      | 0.158<br>(0.312)    |
| GPA                    | 2.322***<br>(0.330) | 0.858***<br>(0.312)  | 0.502*<br>(0.278)   |
| Constant               | 6.952***<br>(0.727) | 6.591***<br>(0.654)  | 5.765***<br>(0.562) |
| Observations           | 1,330               | 1,330                | 1,330               |

*Note:* Standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable in column (1) is the percentage share of “twice” referrals received from the same referrer, in column (2) “single” referral share for cognitive skill, and in column (3) for social skill. Quota is a dummy for the referrals received from classmates in the Quota treatment. Low-SES is a dummy for participant’s socioeconomic status. Remaining independent variables are the respective standardized test scores for skills and GPA.

## Homophily in Social Class

In this section, we explore drivers of the social class bias found in cognitive skill referrals which affect a quarter of all referrals in our sample. Our preregistered hypothesis is that social class bias should be driven by homophily between same-SES classmates, i.e., by the tendency of individuals to form ties with similar others.<sup>9</sup> We can test whether the social class bias in cognitive skill referrals is driven by homophily if we disaggregate referrals according to referrers’ social class. To do so, we decompose  $y_{ic}^{s,single}$  by referrer SES, the same way as in Equation 4.<sup>10</sup> This gives the percentage share of referrals participants get from the two social classes, controlling for the differences in SES distribution across the classrooms. We re-estimate Equation 6 for “single” referrals for cognitive skill across both SES.

Table 7 illustrates our findings. The comparison of interest is the point estimates for participant SES and the effect of the Quota across different low and high SES referrers. In column (1), for “single” cognitive skill referrals coming from low-SES, we find no significant effect of the participant SES or the Quota treatment. Point estimates suggest a slightly lower share of referrals for low-SES in the Baseline, and the Quota treatment seems to decrease this share a bit further. It looks like for low-SES referrers the Quota has no effect at best, or it slightly backfires. Interestingly, we find that low-SES participants rely on the cognitive skill signal when referring instead of the GPA proxy: A one standard deviation increase in cognitive skill score causes significant 0.8 percentage point increase in the share of referrals received, translating to an increase of 18 percent. In column (2), for “single” cognitive skill referrals coming from high-SES, we find a marginally significant effect of the participant SES ( $p = 0.1$ ) with a point estimate that is about two times larger in magnitude compared to the one for low-SES referrers. In the Baseline,

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<sup>9</sup>See McPherson, Smith-Lovin, and Cook (2001) for a discussion on homophily in general, for evidence on social class homophily see Chetty et al. (2022a, 2022b). For experimental evidence on a different type of homophily (gender) in referrals see Beaman, Keleher, and Magruder (2018).

<sup>10</sup>We separate  $\sum_{j \neq i} r_{ijc}^{s,single}$  for referrer treatment and SES. We then divide the sum of referrals by the number of low-SES or high-SES classmates in the treatment applying the same self-referral correction as before.

being low-SES decreases the percentage share of referrals coming from high-SES referrers by 35 percent when controlling for cognitive skill and GPA. The Quota treatment has a significant and an even larger effect on high-SES referral choices, and we find that the entire effect of the Quota is driven by this referrer group. The Quota causes low-SES classmates to get 2.7 percentage points more referrals, a staggering increase of 57 percent. High-SES referrers rely entirely on the GPA proxy when making referrals, and a one standard deviation increase in GPA causes significant 0.8 percentage point increase in the share of referrals received. This is seemingly in contradiction with our earlier hypothesis that the bias should only exist when referring based on the skill where there are differences in performance between social classes. We next explore the possibility that high-SES referrers change the way they refer across treatments, switching from the cognitive skill signal in the Baseline, to the GPA proxy in the Quota treatment.

Table 7: Effects of Quota on Cognitive Skill Referrals by Referrer's SES

|                        | (1)                 | (2)                 |
|------------------------|---------------------|---------------------|
|                        | Low-SES             | High-SES            |
| Quota                  | 0.541<br>(0.754)    | -0.872<br>(0.946)   |
| Low SES                | -0.724<br>(0.717)   | -1.681<br>(0.993)   |
| Quota $\times$ Low-SES | -0.552<br>(0.927)   | 2.711**<br>(1.289)  |
| Cognitive Skill        | 0.805***<br>(0.282) | -0.068<br>(0.354)   |
| GPA                    | 0.334<br>(0.398)    | 0.798*<br>(0.448)   |
| Constant               | 4.359***<br>(0.472) | 4.766***<br>(0.663) |
| Observations           | 1,328               | 1,310               |

*Note:* Standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable is the percentage share of cognitive skill referrals received, split by referrer's SES status. Quota is a dummy for the referrals received from classmates in the Quota treatment. Low-SES is a dummy for participant's socioeconomic status. Remaining independent variables are the standardized test scores for cognitive skill and GPA.

Table 8: Effects of Quota on Cognitive Skill Referrals by Referrer's SES with Skill Interactions

|                                | (1)                 | (2)                 |
|--------------------------------|---------------------|---------------------|
|                                | Low-SES             | High-SES            |
| Quota                          | 0.572<br>(0.775)    | -0.672<br>(0.991)   |
| Low SES                        | -0.711<br>(0.723)   | -1.516<br>(1.016)   |
| Quota $\times$ Low-SES         | -0.580<br>(0.948)   | 2.390*<br>(1.282)   |
| Cognitive Skill                | 0.838*<br>(0.424)   | 0.528<br>(0.381)    |
| Quota $\times$ Cognitive Skill | -0.065<br>(0.412)   | -1.187<br>(0.784)   |
| GPA                            | 0.486<br>(0.439)    | 0.638<br>(0.445)    |
| Quota $\times$ GPA             | -0.303<br>(0.561)   | 0.316<br>(0.655)    |
| Constant                       | 4.344***<br>(0.471) | 4.662***<br>(0.684) |
| Observations                   | 1,328               | 1,310               |

*Note:* Standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable is the percentage share of cognitive skill referrals received, split by referrer's SES status. Quota is a dummy for the referrals received from classmates in the Quota treatment. Low-SES is a dummy for participant's socioeconomic status. Independent variables include standardized test scores for cognitive skill and GPA, as well as their interactions with the treatment dummy.

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## A Additional Figures and Tables

### A.1 Additional Figures

Figure A.1: Stratum distribution of the sample

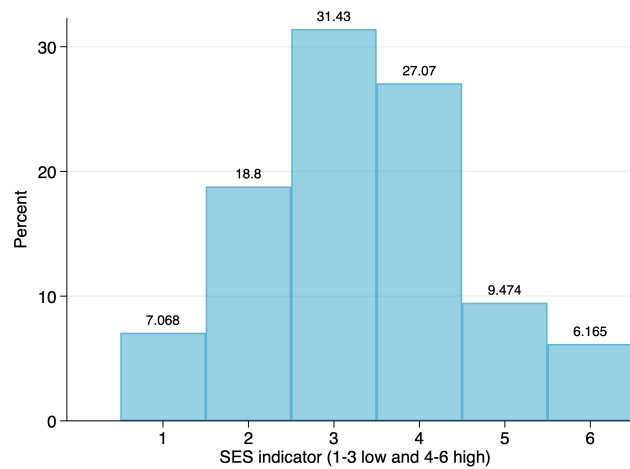
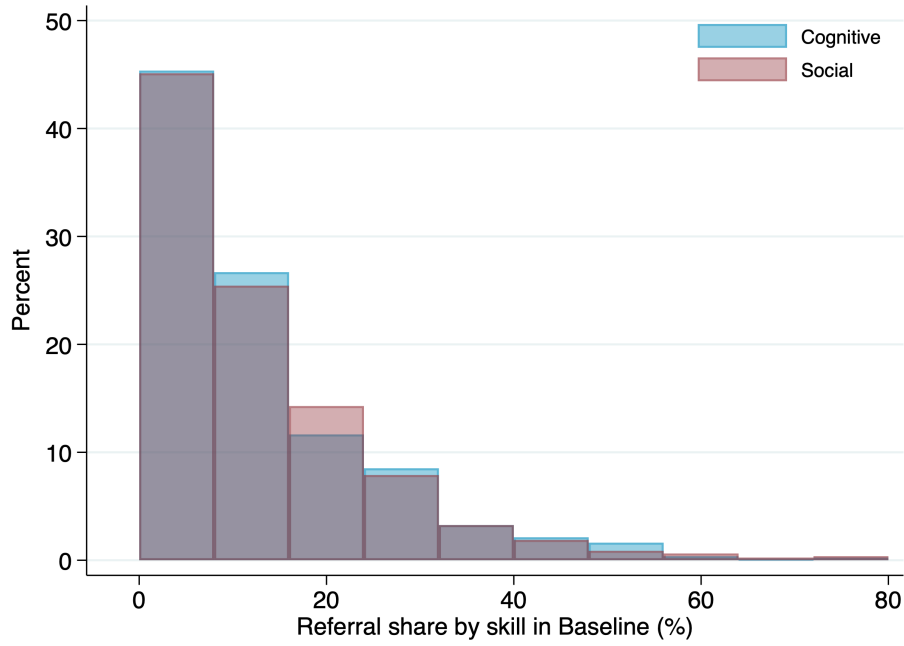
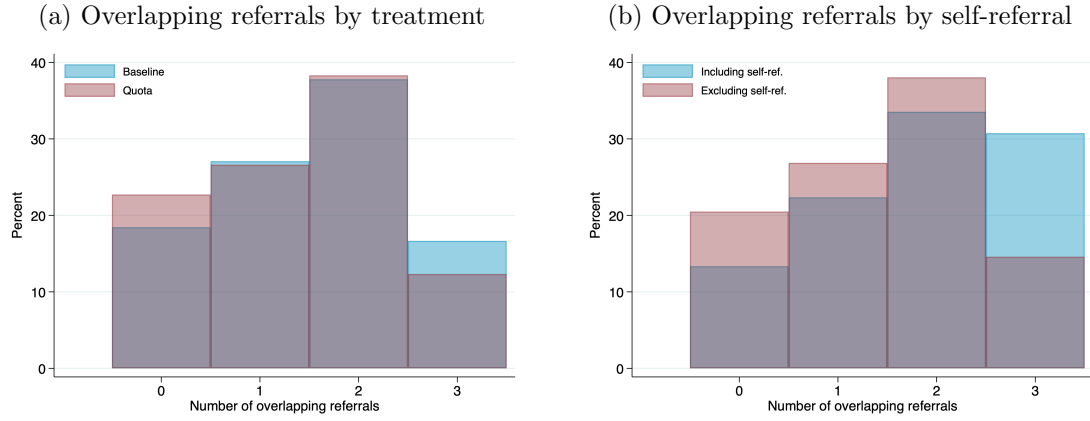


Figure A.2: Referrals in Baseline

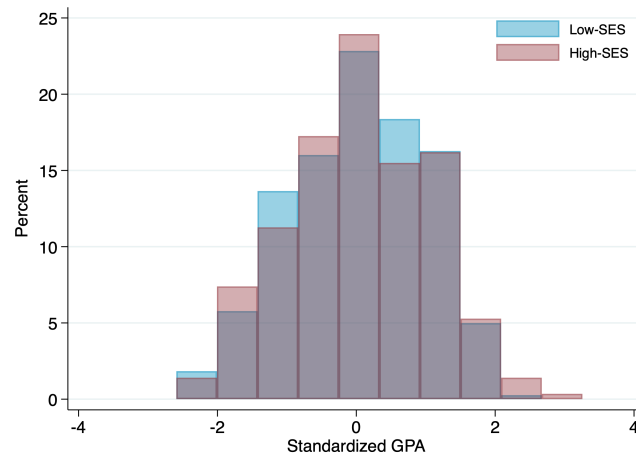


*Note:* The figure shows the percentage share of referrals received across cognitive and social skills from participants in Baseline condition. It is calculated as the number of actual referrals received divided by the maximum possible referrals a student could have received. To illustrate: Consider a classroom with 25 students with 20 participants, and assume 10 participants in each condition. In this classroom, an individual could receive maximum 10 referrals for a skill from classmates in Baseline condition. This number is 9 if the individual is also in Baseline (exclude 1 self-referral possibility). We normalize the referrals received by the maximum to obtain the percentage share shown in the figure. The distribution of referrals is right-skewed with extreme outliers, with consistency across cognitive and social skills.

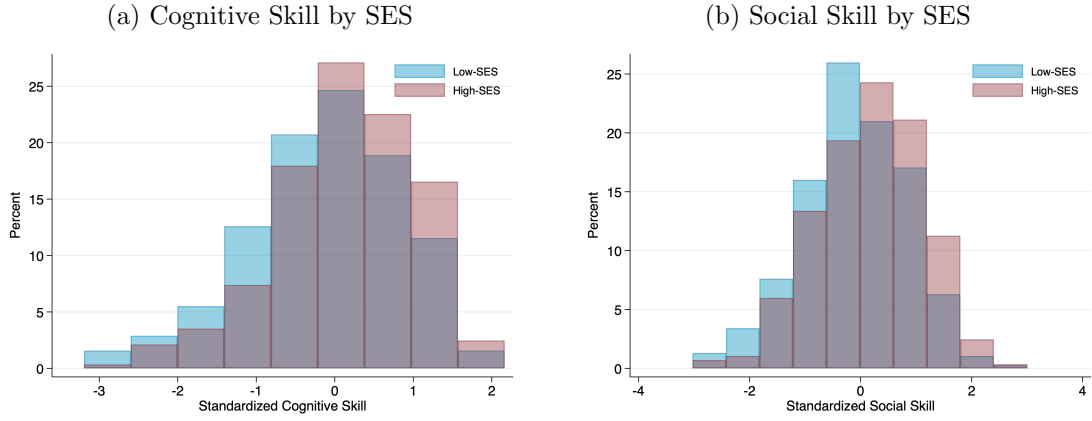


*Note:* Figure (a) shows the distribution of overlapping referrals across treatments, and excludes self-referrals. Figure (b) shows the distribution of overlapping referrals with and without self-referrals. The first bar (value of 0) always indicates the share of participants with 6 unique referrals. The last bar (value of 3) always indicates the share of participants with 3 identical referral choices. We find that treatments do not change referral dispersion, but excluding self-referrals significantly underestimates the share of participants who have 3 identical referral choices. In their original referral choices, about 70 percent of our sample had at least 2 common referrals.

Figure A.4: Distribution of standardized GPA across SES

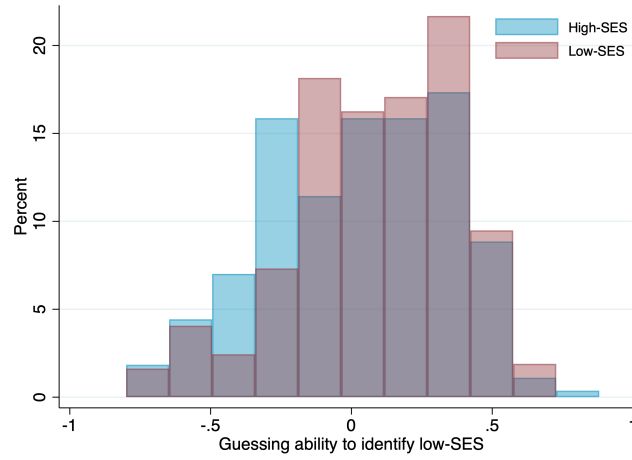


*Note:* Figure shows the distribution of GPA across SES. There are no significant differences in the mean standardized GPA scores between high-SES ( $M = 0.065$ ,  $SD = 1.033$ ,  $n = 284$ ) and low-SES participants ( $M = 0.034$ ,  $SD = 0.971$ ,  $n = 381$ ),  $t(663) = 0.39$ ,  $p = 0.695$ ,  $d = 0.031$ .



*Note:* Figure (a) shows the distribution of cognitive skill across SES. Figure (b) shows the distribution of social skill across SES. For cognitive skill, there are significant differences in the mean standardized scores between high-SES ( $M = 0.177$ ,  $SD = 0.956$ ,  $n = 284$ ) and low-SES participants ( $M = -0.102$ ,  $SD = 1.024$ ,  $n = 381$ ),  $t(663) = 3.58$ ,  $p < 0.001$ ,  $d = 0.279$ . Similarly for mean social skill, there are significant differences between high-SES ( $M = 0.175$ ,  $SD = 0.964$ ,  $n = 284$ ) and low-SES participants ( $M = -0.114$ ,  $SD = 0.989$ ,  $n = 381$ ),  $t(663) = 3.76$ ,  $p < 0.001$ ,  $d = 0.288$ . We can visually verify that high-SES are overrepresented in quantiles above median for both distributions.

Figure A.6: Distribution of guessing ability across SES



*Note:* Figure shows the distribution of the guessing ability across SES. We calculate the guessing ability as the share of succesful low-SES guesses minus the expected probability of randomly drawing low-SES classmates. A score of 0 indicates random draws, below 0 drawing worse than chance, and above 0 better than chance. There are significant differences in the mean guessing ability between high-SES ( $M = 0.022$ ,  $SD = 0.325$ ,  $n = 271$ ) and low-SES participants ( $M = 0.093$ ,  $SD = 0.302$ ,  $n = 369$ ),  $t(638) = -2.85$ ,  $p = 0.005$ ,  $d = 0.226$ . Low-SES participants have significantly higher guessing ability compared to their high-SES counterparts, with a mean difference of 7 percentage points.

## A.2 Additional Tables

Table A.1: Comparison of Variables Between the Sample and Missing Students

|                              | Sample | Missing | <i>p</i> |
|------------------------------|--------|---------|----------|
| Referral share (both skills) | 0.127  | 0.043   | 0.000    |
| GPA (standardized)           | 0.044  | -0.273  | 0.001    |
| Entry Exam (standardized)    | 0.028  | -0.168  | 0.046    |
| # Semesters at UNAB          | 3.171  | 3.188   | 0.884    |
| Age                          | 19.182 | 20.287  | 0.001    |
| Female                       | 49.8%  | 48.5%   | 0.788    |
| Ethnic Minority              | 2.1%   | 4.4%    | 0.114    |
| Rural Community              | 28.8%  | 31.6%   | 0.501    |
| Has Scholarship              | 0.8%   | 0.7%    | 0.899    |

*Note:* Values for female, ethnic minority, rural community, and scholarship represent percentages. All other variables represent means. *p*-values for gender, ethnic, rural, and scholarship are from two-sample tests of proportions. For all other variables, *p*-values are from two-sample t-tests with equal variances. All tests compare the sample and missing students. All reported *p*-values are two-tailed.

Table A.2: Correlation between GPA and skills

|           | GPA    | Cognitive | Social |
|-----------|--------|-----------|--------|
| GPA       | 1.000  |           |        |
| Cognitive | 0.085  | 1.000     |        |
| Social    | 0.093* | 0.267*    | 1.000  |

*Note:* Pairwise correlation between GPA and skills. \* indicates significance at the 5% level with Bonferroni correction. Sample is restricted to 655 participants with complete administrative and experimental data.

Table A.3: Between-Classroom Variation in Academic Programs

| Statistic                 | Most common program share |
|---------------------------|---------------------------|
| Mean                      | 0.424                     |
| Standard Deviation        | 0.216                     |
| 10th percentile           | 0.174                     |
| 25th percentile           | 0.292                     |
| Median                    | 0.345                     |
| 75th percentile           | 0.533                     |
| 90th percentile           | 0.696                     |
| # classrooms with share 1 | 3                         |
| Most diverse classroom    | 0.154                     |
| # classrooms              | 35                        |

*Note:* Table shows the distribution of academic programs across classrooms, measured by the share of students from the most common program in each classroom. Three classrooms are completely homogeneous (share = 1). In the median classroom, the most common program accounts for 34.5% of students. The most diverse classroom has only 15.4% of students in the same program. Data based on 849 students across 35 classrooms.

Table A.4: Characteristics of self-referrers

|                  | No self-referral | Any self-referral | $\Delta$ |
|------------------|------------------|-------------------|----------|
| GPA              | 0.132<br>(1.003) | -0.120<br>(0.966) | 0.252    |
| Cognitive Skills | 0.087<br>(0.988) | -0.118<br>(1.023) | 0.205    |
| Social Skills    | 0.034<br>(1.003) | -0.038<br>(0.959) | 0.072    |
| Low-SES          | 0.605<br>(0.490) | 0.511<br>(0.501)  | 0.094    |
| N                | 440              | 225               | 665      |
| Share (%)        | 66.2             | 33.8              | 100      |

*Note:* Table compares standardized scores between participants who self-referred at least once ( $N = 225$ ) and those who did not ( $N = 440$ ). Positive differences indicate higher scores for those who never self-referred. The results suggest self-referrers have systematically lower cognitive skills and GPA, consistent with the fact that we instructed explicitly against making self-referrals, and are more likely to be low-SES. Standard deviations in parentheses, samples restricted to participants with complete administrative and experimental data.

Table A.5: Characteristics of participants who make overlapping referrals

|                 | All unique referrals | Overlapping referrals | $\Delta$ |
|-----------------|----------------------|-----------------------|----------|
| GPA             | 0.057<br>(0.983)     | 0.045<br>(1.009)      | 0.012    |
| Cognitive Skill | 0.110<br>(1.005)     | 0.024<br>(0.979)      | 0.086    |
| Social Skill    | -0.014<br>(0.938)    | 0.033<br>(0.981)      | -0.047   |
| Low-SES         | 0.530<br>(0.501)     | 0.597<br>(0.491)      | -0.067   |
| N               | 132                  | 512                   | 644      |
| Share (%)       | 20.5                 | 79.5                  | 100      |

*Note:* Table compares characteristics between participants who made overlapping referrals ( $N = 512$ , 79.5%) and those who did not ( $N = 132$ , 20.5%). Overlapping referrals indicate cases where a student referred the same classmate at least once for cognitive or social skills. Positive differences indicate higher scores for those who made no overlapping referrals. The results suggest minimal differences in GPA and social skills. Referrers who made no overlapping referrals have slightly higher cognitive skills and are 6.7 percent more likely to be high-SES. Standard deviations in parentheses, sample restricted to participants with complete administrative and experimental data.

Table A.6: Characteristics of Top Performers and Referrals

|                 | Cognitive        |                  | Social           |                  | Both             |
|-----------------|------------------|------------------|------------------|------------------|------------------|
|                 | Top 3            | Referrals        | Top 3            | Referrals        | Top 3            |
| Cognitive Skill | 1.223<br>(0.419) | 0.112<br>(1.009) | 0.383<br>(0.922) | 0.058<br>(1.015) | 1.201<br>(0.458) |
| Social Skill    | 0.357<br>(0.923) | 0.086<br>(0.996) | 1.340<br>(0.395) | 0.042<br>(1.009) | 1.391<br>(0.453) |
| GPA             | 0.277<br>(0.990) | 0.251<br>(1.021) | 0.264<br>(1.046) | 0.212<br>(1.004) | 0.551<br>(0.897) |
| Low-SES         | 0.457<br>(0.500) | 0.532<br>(0.499) | 0.456<br>(0.500) | 0.555<br>(0.497) | 0.500<br>(0.507) |
| N               | 129              | 1,759            | 114              | 1,775            | 36               |
| Share (%)       | 20.0             | 100              | 17.7             | 100              | 5.6              |

*Note:* Table shows characteristics of students ranked in the top 3 of their classroom and average characteristics of referred students, by skill. Standard deviations in parentheses. Sample restricted to participants with complete administrative and experimental data. All continuous variables are standardized.

## B Experiment

*We include the English version of the instructions used in Qualtrics. Participants saw the Spanish version. Horizontal lines indicate page breaks, and clarifying comments are inside brackets.*

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Please enter the password:

[classroom-specific password sent to each participant the day before data collection]

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## **Welcome**

Welcome to this study organized by the Social Bee Lab. You have been invited to participate in a survey where you can make a series of decisions. The study takes approximately 20 minutes to complete. During the study, you should not communicate with any other students. If you have any questions at any time, please raise your hand. One of the assistants will help you privately.

In this study, you can win bonus money depending on your choices. In total, we will draw [classroom-specific number equal to 40% of class size] bonuses of 100.000 pesos among the participants of this classroom. It is also possible for the same person to win more than one voucher. The following screens will detail how the bonus draw will be conducted. The UNAB finance office will make the payment of the vouchers through Nequi.

All your decisions in this survey will be anonymized. Therefore, the answers you provide will not affect your grades in this class or your records at the university. We will use your personal information to determine the bonus allocation, but after that, we will remove any data that identifies you.

This survey has several parts. Each of these parts has specific instructions. Please read the instructions for each part carefully because they describe how you can earn bonuses. This study has been approved by the [omitted for anonymous review] on the condition that all the information we provide is true and all the bonuses we offer are real. On the next screen, we present you with an informed consent form that you must accept to participate in this study.

---

## **Informed Consent**

You have been invited to participate in a study to learn more about how people make decisions in common scenarios.

This study is conducted by [omitted for anonymous review] and the Social Bee Lab at UNAB. The purpose of this study is to broaden our understanding of how people make decisions.

Participation in this study is voluntary. You may opt-out at any time. No known risks are associated with your participation in this project beyond those of everyday life. Apart from the monetary bonuses that will be drawn, participation has no direct benefits.

The Social Bee Lab is in charge of data collection. Your answers in this study are anonymous and will not be shared with anyone. In addition to your answers, UNAB will provide the Social Bee Lab with administrative records of your courses and your university entrance exam score. Your records, decisions, and your identity will be kept strictly confidential. Data about you collected within the scope of the study are used for scientific purposes only and are treated as strictly confidential. The Social Bee Lab will anonymize your data, and the researcher will analyze it without knowing your identity. All data generated will be stored on the researcher's computer. You have the right to access your personal data and request its deletion. You can exercise this right by contacting the researcher.

If something is unclear or you have any questions, you can contact [omitted for anonymous review].

If you have questions about your rights as a participant, you can contact [omitted for anonymous review].

By continuing to the next screen, you agree to participate in this study.

---

Before you start, please answer these four questions.

What is your gender?

[Male, Female]

What is the socio-economic stratum to which your family belongs?

[Stratum 1 to Stratum 6]

What is your father's highest acquired level of education?

[Primary school, High school, Technical school, Undergraduate, Graduate, Postgraduate, Not applicable.]

What is your mother's highest acquired level of education?

[Primary school, High school, Technical school, Undergraduate, Graduate, Postgraduate, Not applicable.]

---

## **Part 1**

You will now participate in two quizzes, each lasting five minutes. Please try to answer

them to the best of your ability.

We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 100.000 pesos in this first part. The steps to allocate the bonuses for Part 1 are explained below.

---

[classroom-specific illustrations explaining the incentive structure]

---

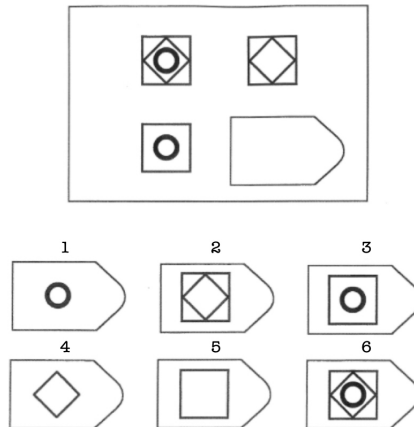
[random assignment to either cognitive or social skills test]

### **Test - Cognitive Skill**

In this test, you will see a series of images. Below is an example of the images you will solve. At the top of each image, there is a pattern with a piece that has been removed. Your task is to choose which of the six pieces completes the pattern correctly. For each image, there is only one correct piece. Look at the following example:

First, notice a square in the upper left, the upper right, and the lower left. Also, notice that the circle is eliminated when one moves from the upper left to the upper right. Finally, the rhombus is eliminated when moving from the upper left to the lower left. Therefore, the correct piece should eliminate the circle and the rhombus, leaving only a square. So, the correct answer is piece 5.

To give your answer to each image, you must choose the correct option and then continue to the next screen. After giving your answer you cannot go back.



You will have 5 minutes to complete the test, which consists of 18 images to solve. The percentage of correct answers will determine your chances of winning one of the 100.000 pesos bonuses if you are chosen for the drawing.

---

Are you ready?

Your 5 minutes will start as soon as you move to the next screen.

---

### Problem 1

[screenshot of Raven's matrix]

[After participants submit an answer, a new matrix appears on the screen. The sequence of matrices is the same for all participants. Participants cannot return to a

previous screen. Participants do not have to provide answers for all 18 matrices.]

---

You have finished the test. You can proceed to the next screen.

---

### **How did you do on the test?**

If we randomly choose 10 participants from this classroom, how many people do you think solved fewer correct problems than you?

[Slider from 0 to 10]

---

### **Test - Emotions**

In this test, you will see a series of photographs. Below is an example of the pictures you will see. In each picture, you will see the eyes of a person. Below the picture, you will see four possible emotions that this person is feeling. Your task is to choose which of the four emotions correctly describes what the person is feeling. For each picture, there is only one emotion. Look at the following example:



[Happy, Disappointed, Shocked, Worried]

In this case, the correct answer is: Shocked.

To give your answer to each picture, you must choose the correct option and then continue to the next screen. After giving your answer you will not be able to go back.

You will have 5 minutes to complete the test, which consists of 36 photographs to solve. The percentage of correct answers you get will determine your chances of winning one of the 100.000 pesos bonuses if you are chosen for the drawing.

---

Are you ready?

Your 5 minutes will start as soon as you move to the next screen.

---

### **Photograph 1: Choose the word that best describes the photograph**

[photo from Multiracial Reading the Mind in the Eyes Test]

[After participants submit an answer, a new photo appears on the screen. The sequence of photos is the same for all participants. Participants cannot return to a previous screen. Participants do not have to provide answers for all 36 photos.]

---

You have finished the test. You can proceed to the next screen.

---

### **How did you do on the test?**

If we randomly choose 10 participants from this classroom, how many people do you think solved fewer correct photographs than you?

[Slider from 0 to 10]

---

### **Part 2**

At the beginning of this study, all participants took two tests, one on cognitive ability and one on emotions. In this part, we will ask you to recommend the people who in your opinion will score the best on each test.

You may recommend 3 people per test, but you may not recommend yourself.

We will allocate up to [classroom-specific number equal to 40% of class size] bonuses of 100.000 pesos for Part 2. The steps for allocating bonuses are explained below.

---

[random assignment to either quota or baseline condition]

[classroom-specific illustrations explaining the incentive structure depending on assign-

ment to either baseline or quota conditions]

Figure A.7: Illustrations for the two conditions



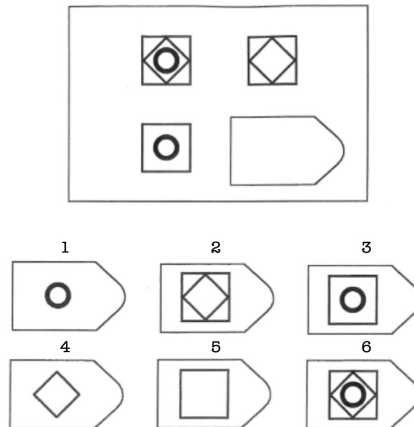
[random assignment to either cognitive or social skills referral task]

### Recommendation - Cognitive Skill

All participants took a test to identify the missing pattern in each image, as in the example below. This test is used to measure general intelligence.

Next, we will present you with a list of the names of all the students in this room. We will ask you to recommend the three people you think will score the highest on the general intelligence test.

If you are chosen by the computer, each of your recommendations in the top 3 increases your chances of winning one of the 100.000 pesos bonuses.



Select the students in this classroom who you consider to have the highest scores on the general intelligence test. (Select 3 students)

[Classroom-specific list of all classmate names visible on one screen. Participants have to pick 3 classmates to continue. Picking their own name invalidates their choices.]

---

### Recommendation - Emotions

All participants took a test where they had to identify the emotion that best described the expression of each image as in the example below. This test is used to measure social skills.



Next, we will present you with a list of the names of all the students in this room. We will ask you to recommend 3 people you think will score the highest on the social skills test.

If you are chosen by the computer, each of your recommendations in the top 3 increases your chances of winning one of the 100.000 pesos bonuses.

---

Select the students in this classroom who you consider to have the highest scores on the social skills test. (Select 3 students)

[Classroom-specific list of all the names visible on one screen. Participants have to pick 3 classmates to continue. Picking their own name invalidates their choices.]

---

### **Part 3: Recommendation - Random draw**

In this part, the computer will randomly choose three students who belong to strata 1, 2, or 3. We will ask you to nominate three people you think the computer will choose.

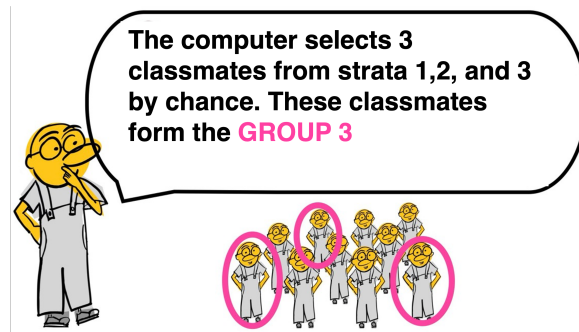
We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 100.000 pesos for Part 3. The steps for allocating the bonuses are explained below.

---

[classroom-specific illustrations explaining the incentive structure]

---

Figure A.8: Illustration for the Guessing Task



Select the students in this classroom who belong to strata 1, 2, or 3, who you think will be randomly selected by the computer (Select 3 students).

[Classroom-specific list of all the names visible on one screen. Participants have to pick 3 classmates to continue. Picking their own name invalidates their choices.]

---

#### Part 4

Do you want to know your scores on the general intelligence test and the social skills test? We can analyze the data and give you a report that explains your strengths in these two areas. Also, what do these strengths mean, and how can you leverage them for your personal and professional development?

If you want to receive your skills report, we need to contact you again. We also want to be able to invite you to new studies where you can participate for more bonus money. Please indicate if you agree to be contacted again.

[I can be contacted for new studies and to send me my report. I can be contacted to

send my report, but not for new studies. No, I do not want to be contacted again.]

---

[if participant gives consent to be contacted again]

Please enter your contact email:

[student email]